GLT-STEER: Transformation Deltas as Activation-Space Editors for Final Markers

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Abstract

This short draft studies whether linguistic transformation vectors learned from sentence-pair hidden-state differences can be injected back into a generative transformer as activation-space editors. In GPT-2, vectors derived from controlled transformation pairs reliably steer final-position surface markers such as ?, !, and ... under matched controls. The effect is visible in generated text, next-token marker logits, position-of-intervention audits, and a fixed-parameter confirmatory rerun on fresh hard-heldout sources. The claim is intentionally bounded: current GLT-STEER supports final-marker form steering, not general semantic rewriting and not a Lie-algebra claim.

1. Question and Scope

For a source sentence x and transformed sentence y, GLT computes a hidden-state displacement delta = h(y) - h(x). Earlier GLT tracks test whether these deltas classify transformation type offline. GLT-STEER asks a stronger intervention question: if a transformation delta is injected during generation, does the model produce the corresponding transformation marker?

The current defensible claim is that transformation deltas can act as activation-space editors for final-position surface markers in GPT-style residual streams. The cleanest evidence is in GPT-2, with marker-dependent transfer to DistilGPT-2. This is not a claim of robust semantic rewriting.

2. Method

- Build synthetic source/target pairs for a transformation class.

- Extract hidden states at selected GPT-style residual blocks.

- Compute a centroid transformation vector from training pairs.

- During generation, add the vector to the residual stream.

- Compare target steering against no-steering, wrong-marker, random-norm, and negative-vector controls.

```
delta = h(target sentence) - h(source sentence)

y_pred generation: residual[layer] += gain * centroid_delta
```

3. Main Evidence

3.1 Question steering and prompt-only control

condition	question mark rate
target question vector	0.9350
random-norm control	0.0000
wrong-class control	0.0000
negative-target control	0.0000

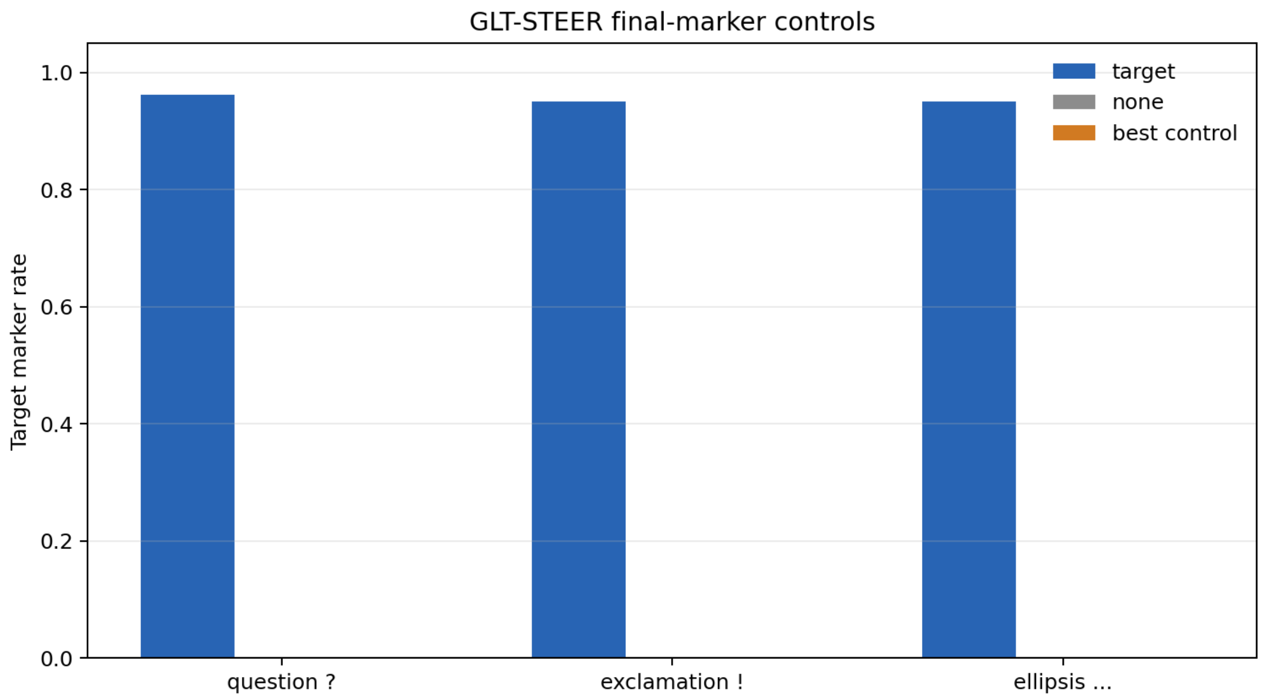
Copy-like prompts without steering produce 0.0000 question marks across 960 no-steering rows from GPT-2 and DistilGPT-2. This rules out the simplest prompt-only explanation for the question-marker result.

3.2 Content preservation

source set	prompt family	target question-and-preserved
in-template	copy-like prompts	0.9625-0.9750
simple out-of-template	copy-like prompts	0.8250-0.9000

Hard out-of-template sources reduce preservation. GPT-2 still shows question-form steering; DistilGPT-2 is weaker and should be reported as marker-form transfer only.

Figure 1. Final-marker controls



Exclamation and ellipsis controls show that the question result is part of a broader final-marker phenomenon, not only a question-specific artifact.

4. Mechanistic Audits

4.1 Logit-level final-marker audit

target	control	marker rate (N, 95% CI)	median best rank
?	none	0.0000 (N=96, [0.0000, 0.0385])	28.0
?	target	0.8542 (N=96, [0.7700, 0.9111])	1.0
!	none	0.0000 (N=96, [0.0000, 0.0385])	20.0
!	target	0.9063 (N=96, [0.8313, 0.9499])	1.0
...	none	0.0000 (N=96, [0.0000, 0.0385])	11.75
...	target	0.8750 (N=96, [0.7941, 0.9270])	1.0

No-steering marker rates are zero, while target steering moves the intended marker token to rank 1 in most sequences. This is the strongest current evidence that the intervention changes the generation distribution itself.

4.2 Position-of-intervention audit

condition	position mode	question rate	question+preserved
none	none	0.0000	0.0000
target_last_each_step	last_each_step	0.9604	0.7917
target_prompt_all_once	prompt_all	0.8625	0.7896
target_prompt_first	prompt_first	0.0000	0.0000
target_prompt_middle	prompt_middle	0.0000	0.0000
target_prompt_last_once	prompt_last	0.0000	0.0000

Single prompt-position edits do not work. Distributed prompt-state editing or repeated last-token-at-each-step editing does work. This argues against a trivial one-token prompt perturbation story.

5. Fixed-Parameter Confirmation

model	target	target marker rate	max control marker	target marker+preserved	max control marker+preserved
distilgpt2	ellipsis	0.8333 (N=144, CI [0.764, 0.885])	0.0000	0.1597 (N=144, CI [0.109, 0.228])	0.0000
distilgpt2	exclamation	0.8958 (N=144, CI [0.835, 0.936])	0.0139	0.3472 (N=144, CI [0.274, 0.428])	0.0000
distilgpt2	question	0.6319 (N=144, CI [0.551, 0.706])	0.0069	0.0556 (N=144, CI [0.028, 0.106])	0.0000
gpt2	ellipsis	0.8438 (N=288, CI [0.797, 0.881])	0.0000	0.3854 (N=288, CI [0.331, 0.443])	0.0000
gpt2	exclamation	0.6875 (N=288, CI [0.632, 0.738])	0.0000	0.3958 (N=288, CI [0.341, 0.453])	0.0000
gpt2	question	0.6562 (N=288, CI [0.600, 0.709])	0.0000	0.2396 (N=288, CI [0.194, 0.292])	0.0000

This run uses fresh hard-heldout sources and fixed settings: GPT-2 layers 2,3 at gain 0.75; DistilGPT-2 layer 2 at gain 1.0. No layer/gain search is performed inside the run. Target marker rates remain separated from controls across all three final markers and both tested models.

6. Boundary Results

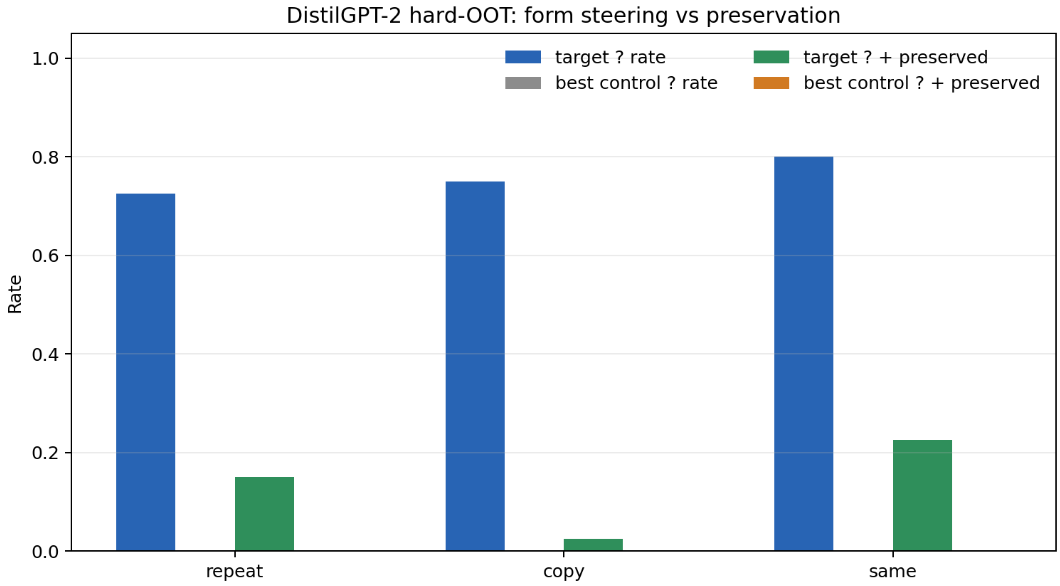
6.1 Negation

Negation is weak under the same copy-prompt recipe. A GPT-2 layer sweep over layers 0-11 does not find a clean intervention site. The best target-and-preserved row reaches only 0.1729, while matched controls remain nontrivial, with maxima up to 0.1229.

layer	question mean pairwise cosine	negation mean pairwise cosine
2	0.9693	0.5621
3	0.9365	0.5665

6.2 DistilGPT-2

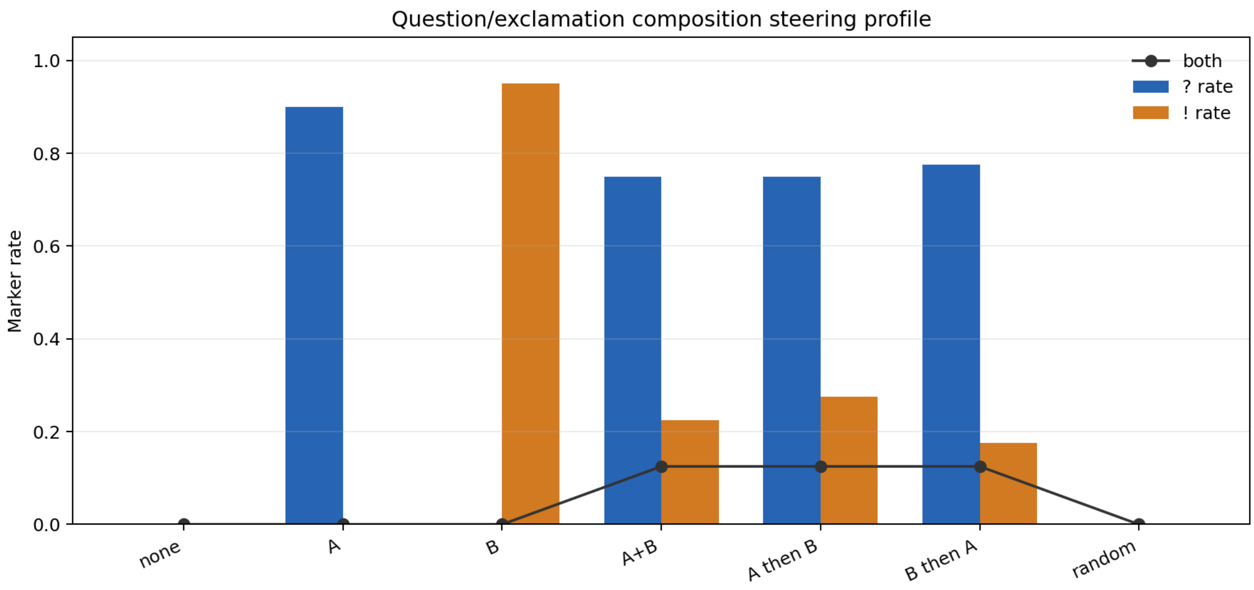
Figure 2. DistilGPT-2 hard-OOT boundary



DistilGPT-2 preserves marker-form steering but has weak strict content preservation on hard out-of-template sources.

6.3 Marker composition

Figure 3. Marker composition



Combined question/exclamation vectors produce mixed marker profiles and lower preservation. The correct interpretation is competition/saturation, not noncommutative algebra.

7. Related Work Positioning

GLT-STEER is closest to activation-engineering and representation-steering work. Activation Addition computes activation differences from contrastive prompts and injects them at inference time. Contrastive Activation Addition averages residual-stream differences over positive and negative behavioral examples. Representation Engineering frames a broader family of methods for monitoring and manipulating model internals.

The contribution here is not that activation addition exists. The narrower contribution is diagnostic: GLT-STEER derives vectors from controlled linguistic transformation pairs and asks which linguistic transformations behave as reusable intervention vectors. The current answer is bounded: final-position markers are steerable; sentence-internal transformations are not solved by this recipe.

8. Limitations

- GLT-STEER does not solve semantic rewriting.

- Negation and modality are not cleanly steered by the current method.

- Final-marker steering does not prove general linguistic transformation editing.

- Marker composition does not prove noncommutative algebra.

- The result does not establish a Lie algebra in transformer activations.

- Prompt wording remains a real boundary condition.

9. Reproducibility

All reported results are archived in results/experiments/. No new model inference is required to inspect the tables in this draft. The datasets are synthetic controlled sentence-pair templates generated by repository scripts.

artifact	path
draft source	paper/articles/glt-steer-activation-editors/submission_draft.md
PDF builder	scripts/build_glt_steer_submission_pdf.py
generated PDF	reports/2026-08-25_glt_steer_submission_draft.pdf
confirmatory result	results/experiments/glt_steer_confirmatory_fixed_params_20260825_results/
CI audit	results/experiments/glt_steer_headline_ci_20260825_results/

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